



MODULE NAME:	MODULE CODE:
PROGRAMMING 3D	PROG7314
OPEN SOURCE CODING (INTERMEDIATE)	OPSC7312

ASSESSMENT TYPE: POE (PAPER AND MARKING RUBRIC)

TOTAL MARK ALLOCATION: 300 MARKS

TOTAL HOURS: A MINIMUM OF 35 HOURS IS SUGGESTED TO COMPLETE THIS ASSESSMENT

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Please ensure that you submit your assignment through Turnitin.***
3. ***The assignment must be pushed to GitHub.***
4. ***Assignments must be typed unless otherwise specified.***
5. ***Follow all instructions on the PoE cover sheet.***
6. ***This is a group assessment. Please allow for a maximum of 4 participants.***

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where **both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g. alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruities: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruities: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

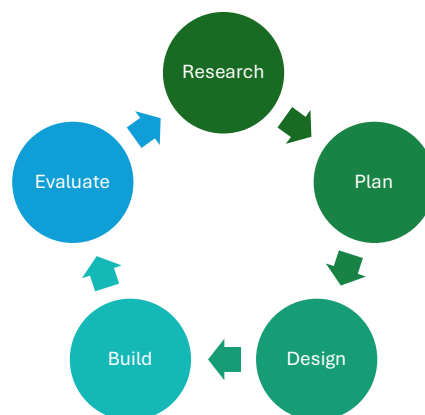
Background

This portfolio of evidence comprises two parts submitted during the semester, and the finalisation of your POE, each of which takes you through conceptualising, designing, and building a mobile application.

You are going to create an app of **your choice** for a change. The app must make use of a hosted **API** that you create yourself, utilising the knowledge learnt in PROG7311, PROG7312, CLDV6212/APPR6212, and/or INSY7314/APDS7312. The app must include the stated features described in the introduction section below, and the API must be responsible for most of the heavy lifting of your app.

Finally, the app and related assets will need to be prepared so that it is ready to be published on the Google Play Store, as well as making use of unit testing and GitHub Actions for running the necessary tests.

To be able to create the app, you will need to use all the techniques you learned during this course, as well as conduct additional research on how to complete some of the tasks required of you. It is also critical that you approach the project in a methodical and planned manner. For this reason, the project will follow a Research – Plan – Design – Build – Evaluate cycle:



This POE will run for one full cycle. After one full cycle, you can always restart the process to refine your app further.

The different phases of the cycle will be related to the different tasks as follows:

- **Part 1: Research, Planning and Design**
- **Part 2: Build**
- **Final POE: Build and Evaluate.**

Introduction

Each of the below parts builds on top of the other. Be sure to complete or update one part before moving on to the next. Each part will be individually assessed by your lecturer, and feedback for each part will be provided.

The exact features, design, and layout of your app are up to you, but each app must be able to complete at least the following features:

- The user must be able to **register** and **log in** to the app using single sign-on (**SSO**).
- Make use of **Biometric Authentication**: Implement fingerprint or facial recognition for secure user authentication. **(POE only)**.
- The user must be able to change their **settings** in the app.
- Connect to a **REST API** you create that is connected to a database.
- **Offline mode** with Sync: Create a feature that allows users to perform actions offline with synchronisation capabilities once they reconnect. This does not need to be the whole app, but at least some features. You can make use of RoomDB or SQLite for the offline storage **(POE only)**.
- Implementation of **Real-time Notification**: Implement a push notification system for real-time updates and alerts **(POE only)**.
- **Multi-language** support that includes at least two South African Languages **(POE only)**.

These features are the **minimum** that's required. Your app also needs to work **logically** and be **easy to use**. Speak to your lecturer about how your app can best implement all these features.

Part 1 — Research, Planning and Design**(Marks: 100)**

At the end of this specific part, students should be able to:

- Explain how to connect to a RESTful Application Programming Interface (API).
- Define the features that the app will include.
- Define the use case of their app.

This part is composed of two sections, weighted equally:

- Research
- Planning and design

Research**(Marks: 50)**

For this part, you are required to research **three** existing applications available on the Android operating system that are functionally similar to the application you intend to create. You will need to present your analytical findings in the form of a professional, technically rigorous research report.

- **Word Count:** 1500 to 2000 words.
- **Format:** Typed PDF document.
- **File Constraints:** Ensure the final PDF file size does not exceed 10 MB.

Note: If you cannot find an application that matches your exact concept, evaluate applications that implement at least some of your planned core features or targeted system functionalities.

Your research report must be structured using the following explicit sections:

- **Introduction:** Outline the purpose of your research, establish the domain of your proposed app, and preview the sections of the document.
- **Competitive Analysis (Repeated for App 1, App 2, and App 3):**
 - **Overview of the App:** A brief summary of the application's core purpose, primary target audience, and main use case.
 - **Strengths and Weaknesses:** A critical evaluation of the application's functional advantages and current user experience limitations.
 - **Technical Implementation Theory:** Analyse how the app was built based on your knowledge of modern native Android development frameworks. Hypothesise their software architectural choices (such as MVVM or MVI patterns), state management approaches, data persistence layers, and UI composition strategies (such as Jetpack Compose versus XML).

- **Interface Architecture:** Provide high-quality screenshots illustrating critical user interface flows and layout structures.
- **Comparison Matrix:** Present a comprehensive visual comparison (such as a structured comparison table) map out the key similarities, design differences, and functional variations across all three applications.
- **Feature Adoption Strategy:** A prioritised list of the best functional or structural features observed across the researched apps, complete with a technical justification explaining why you want to use them in your final application.
- **Conclusion:** Conclude the report by summarising your findings and explicitly detailing how this research directly establishes the foundation for your upcoming Planning and Design document.
- **References:** Ensure all software, documentation, and external academic resources are fully and consistently cited using the correct technical referencing style.

Planning and Design

(Marks: 50)

For this part, you will design your mobile application completely. The primary goal of this phase is to construct a comprehensive technical blueprint and system architecture specification before you begin writing code for the prototype in Part 2.

- **Word Count:** 2000 to 2500 words.
- **Format:** Typed PDF document.

Your Planning and Design document must be structured using the following explicit sections:

- **Introduction:** Set the context for your project, defining the high-level scope of the application and outlining the structure of this design specification.
- **Application Overview:** Provide a concise profile of your application concept, including:
 - The official name of the app.
 - An initial application icon design asset.
 - A descriptive breakdown of the innovative features you plan to build (such as gamification elements).
- **Detailed Requirements Specification:** Compile an exhaustive list of system requirements. You must elaborate on every minimum core requirement specified in this assignment brief instead of simply copying and pasting the text. Factor in your own custom features discovered during research. The requirements must be granular and explicit enough that an external developer could build your exact application solely by reading this document.

- **User Interface (UI) and Navigation Design:** Map out the visual layout and user journeys:
 - A clear user interface mockup for every single screen.
 - A functional description explaining the exact purpose and interactive elements of each screen.
 - A navigation wireflow diagram illustrating exactly how a user transitions between the various screens.
- **REST API and Back-End Architecture:** Define your custom web service:
 - A structural plan detailing what your API will handle for the application.
 - An endpoint specification detailing the exact data schemas (payloads) it will send and receive.
 - The development stack you will use to create the API and your selected cloud hosting platform. Remember, you can use credits you get from GitHub Education for this (Azure for Students, Digital Ocean, etc.)
- **UML System Architecture Diagrams:** Include Unified Modelling Language (UML) diagrams that visually demonstrate how the mobile application interacts with your custom REST API, databases, and any other external software development kits (SDKs) or systems.
- **Data Models and Schema Definitions:** Provide an exhaustive listing of the data that the app needs to capture from the user and persist. You must specify the explicit data types (such as String, Integer, Boolean, or Custom Objects) for every field.
- **Project Management Plan (Gantt Chart):** Outline your implementation timeline using a visual Gantt chart. Break milestones down into highly specific, actionable engineering tasks rather than vague descriptions like "implement prototype". You must explicitly allocate dedicated time blocks for software testing and bug fixing cycles.
- **Conclusion:** Summarise your design choices and confirm your structural readiness to pivot into the implementation phase.
- **References:** List all external design frameworks, architectural patterns, and engineering documentation utilised using a consistent referencing standard.

Part 1 Submission Checklist:

To successfully complete and submit Part 1, you must upload the following three separate files:

1. **Research Report:** Your completed comparative analysis report submitted in a clean PDF format.
2. **Planning and Design Document:** Your finished technical blueprint document submitted in a clean PDF format.

3. **AI Usage Write-Up (If Applicable):** A short, transparent narrative (maximum 500 words) detailing exactly how generative AI tools were utilised and cited during this phase. This includes assistance with generating code architecture snippets, troubleshooting design errors, creating initial mockup concepts, or generating asset placeholders.

Part 2 – App Prototype Development

(Marks: 100)

At the end of this specific part, students should be able to:

- *Use a RESTful API within a native Android application.*
- *Integrate external libraries to solve specific programming challenges.*
- *Connect your mobile application to an appropriate Software Development Kit (SDK).*
- *Design and execute detailed software unit testing.*

Prototype Requirements and Scope

For this submission, you must build a fully functional application prototype based on your design specifications from Part 1.

- **Compilation:** Your application must successfully compile and run on a physical device for marks to be awarded for its features.
- **Feature Scope:** Incorporate the core minimum requirements detailed below, excluding features explicitly designated for the final POE phase. You must include at least a few of your custom user-defined features from your design document.
- **Visual Assets:** Final production assets (such as finalised logos or high-fidelity images) are not mandatory at this stage and can be deferred to the final POE.
- **Application Robustness:** The user interface must be intuitive, user-friendly, and capable of gracefully handling invalid inputs through validation checks without crashing. The prototype should operate with only minor bugs or structural errors.

Technical Deliverables and GitHub Integration

1. Source Code and Logging

- Implement the entire application using native Kotlin source code.
- Include clear comments throughout your codebase and provide proper technical referencing where external code blocks or patterns were adopted.
- Incorporate functional logging (using standard Android Log utilities) to demonstrate a clear programmatic understanding of your application's lifecycle and state transitions.

2. Repository Management and DevOps

- Initialise a GitHub repository containing a comprehensive README.md file. **ZIP file submissions or manual file uploads are strictly prohibited.**
- Establish a professional version control workflow by committing and pushing your changes to the repository regularly as you develop features.
- Configure automated testing and continuous integration using GitHub Actions. Your workflow must trigger automated builds and execute your unit tests on every push to verify your codebase functions outside of your local environment.
- You can utilise the following workflow guides to construct your automation file:
 - [GitHub Actions Android Build Marketplace Guide](#)
 - [Reference Build Workflow Configuration](#)

3. Repository Documentation

Your README.md file serves as your core technical report for this phase. Customise it with markdown formatting, architectural diagrams, and embedded screenshots to detail:

- The core purpose and scope of the mobile application.
- Structural design considerations and architectural choices.
- A comprehensive summary explaining your implementation of version control and GitHub Actions automation workflows.

Video Demonstration Requirements

You must record a professional, high-quality demonstration video of the prototype executing live on a physical mobile phone. The video must include an explanatory voice-over and cover the following functionalities:

- **Authentication:** A user registering and logging into the application successfully utilising Single Sign-On (SSO) systems.
- **Application State:** Changing and saving user preferences via a dedicated settings menu.
- **Web Services:** Successful data round-trips connecting to your custom, cloud-hosted REST API.
- **Data Verification:** Visual confirmation showing exactly how data is being captured and modified inside your live cloud authentication service, your custom API layer, and your linked database.
- **Custom Scope:** A demonstration of the core user-defined features you specified during your Part 1 design phase.

Part 2 Submission Checklist

Ensure that your final digital submission includes access to the following items:

- **GitHub Repository Link:** Direct access to your complete Kotlin project source code, active workflow runs, and formatted documentation.
- **Demonstration Video Link:** An unlisted YouTube link or accessible video URL embedded directly within your repository's README file.
- **AI Sub-Report (If Applicable):** A brief summary write-up (not exceeding 500 words) detailing the exact ways generative AI frameworks were used to debug errors, structure files, or optimise your prototype code

You can upload your video to YouTube, make it unlisted, and add that link to your readme file.

You can also use any video recording software you like. Just make sure to include a voice-over to show your understanding.

POE — Final Asset Development, Additions, Tweaking and Final Submission (Marks: 100)

At the end of this specific part, students should be able to:

- *Use blob storage.*
- *Use a NoSQL database to store data.*
- *Use a NoSQL database to read data.*
- *Prepare an app for publication in the Play Store.*

The final application must run natively on a physical mobile device rather than an emulator. Students must plan well in advance to secure an Android device if they do not own one personally. For this final submission, the application must be a fully functional, error-free product that incorporates prior feedback provided by the lecturer throughout the semester. The application must successfully compile and run for marks to be awarded for its technical features. The user interface must remain user-friendly, appropriate, and robust enough to handle invalid user inputs gracefully without experiencing runtime crashes.

Advanced Feature Requirements (POE Only)

In addition to the core prototype features, the final application must implement the following advanced technical capabilities:

- **Biometric Authentication:** Implement secure user authentication using native fingerprint or facial recognition frameworks.

- **Offline Mode with Synchronisation:** Create a resilient data architecture (utilising RoomDB or SQLite) that allows users to perform actions offline and automatically synchronises data once a network connection is re-established. This capability can apply to specific target features rather than the entire application.
- **Real-time Notifications:** Implement a live push notification system using Firebase Cloud Messaging to send real-time updates and alerts.
- **Multi-Language Support:** Integrate local accessibility support that includes at least two official South African languages.
- **Production Assets:** Configure a custom application icon and finalise all visual user interface image assets.

Developer Documentation Resources: You may consult external engineering documentation to implement biometrics and push notifications. The following official resources are provided to assist you:

- Biometric Authentication: [Android Biometric Codelab Guide](#) and [Android Biometric Identity API Reference](#)
- Push Notifications: [Firebase Cloud Messaging Documentation](#)

Technical Architecture and DevOps

1. Version Control and Release Management

- Initialise the repository with a cleanly formatted README.md file.
- Maintain an active and clear version control workflow by committing and pushing code changes regularly as progress is made.
- **Critical Step:** Since this is the final submission, you must explicitly tag your final deployment commit as the Final POE.

2. Automated CI/CD Workflows

- Utilise GitHub Actions to handle automated builds and run your suite of unit tests on every remote push.
- This automated verification ensures your source code compiles cleanly outside your local machine environment.
- You can reference the following implementation guides to construct your workflow file:
 - [Automated Build Android App Marketplace Action](#)
 - [Sample Build Configuration Workflow Script](#)

Comprehensive Demonstration Video

You must record a professional video presentation demonstrating your completed application running live on a physical mobile phone. The video must include an explanatory voice-over narrative and clearly showcase the following verification points:

- **Authentication Flows:** A user registering and logging into the application successfully utilising Single Sign-On (SSO) systems alongside Biometric verification.
- **Application Settings:** Modifying and saving user preferences within a dedicated settings menu layout.
- **Data Pipeline Sync:** Real-time data communication with your custom cloud-hosted REST API.
- **Offline Architecture:** Simulating network disconnection, executing an action in offline mode, and demonstrating successful data synchronisation upon reconnection.
- **Cloud and DB Infrastructure:** Visual confirmation of real-time data adjustments occurring within the hosted authentication service, custom API endpoints, and the back-end database.
- **Core Deliverables:** A clear presentation of your user-defined features specified during the Part 1 design phase.

POE Submission Checklist

Your final submission on GitHub must be clean, organised, and contain no compressed zip files.

Ensure the following artefacts are accessible:

- **Kotlin Source Code:** The complete source code for the Android application submitted directly to your repository. All code blocks must contain professional comments and clear technical referencing where applicable.
- **System Logging:** Active use of Android logging utilities throughout the codebase to showcase an understanding of app execution lifecycles and state changes.
- **Technical Readme:** A detailed README.md file outlining the application's purpose, design considerations, and DevOps integrations. Customise it with images and structure it as an official technical report.
- **Demonstration Video Link:** A working URL or unlisted YouTube link to your presentation video embedded directly within the project README file.
- **Formal Release Notes:** A dedicated section or document detailing every functional update made to the application since the prototype phase. The notes must clearly explain which of your innovative features were added to this final version.

- **Store Publication Readiness:** Verifiable evidence of preparation for the Google Play Store. This must include a signed production APK and promotional screenshots.
- **AI Write-Up (If Applicable):** A transparent summary narrative (maximum 500 words) defining how generative AI tooling was utilised and cited to support code generation, bug fixing, error resolution, or asset generation.

You can upload your video to YouTube, make it unlisted, and add that link to your readme file. You can also use any video recording software you like. Just make sure to include a voice-over to show your understanding.

Assessment Sheet (Marking Rubric)

Please note: Tear off this section and **attach** it to your work when you submit it/ If this is an online submission, then this information needs to be included in the online submission.

MODULE NAME:	MODULE CODE:
PROGRAMMING 3D	PROG7314
OPEN SOURCE CODING (INTERMEDIATE)	OPSC7312

STUDENT NAME:
STUDENT NUMBER:

PART 1 – Research				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard.
Research: Introduction [5 Marks]	- No introduction included. - The introduction does not link to the rest of the document.	Acceptable and links to the rest of the document, but does not explain the purpose of the research.	Acceptable and links to the rest of the document with some details about the purpose of the research.	An excellent introduction that explains the purpose of the research and the sections included in the document.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks
Research: App 1 research [8 Marks]	- Not included or sections missing. - All sections are included, but limited details are provided.	All sections are included, but more details could be added to most of the sections.	All sections are included, but more details could be added to some of the sections.	Comprehensive discussion differentiating between the strengths and weaknesses and motivating why features are considered innovative.
	0 – 1 Marks	4 – 5 Marks	6 Marks	7-8 Marks
Research: App 2 research [8 Marks]	- Not included or sections missing. - All sections are included, but limited details are provided.	All sections are included, but more details could be added to most of the sections.	All sections are included, but more details could be added to some of the sections.	Comprehensive discussion differentiating between the strengths and weaknesses and

PART 1 – Research				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard.
				motivating why features are considered innovative.
	0 – 1 Marks	4– 5 Marks	6 Marks	7-8 Marks
Research: App 3 research [8 Marks]	- Not included or sections are missing. - All sections are included, but limited details are provided.	All sections are included, but more details could be added to most of the sections.	All sections are included, but more details could be added to some of the sections.	Comprehensive discussion differentiating between the strengths and weaknesses and motivating why features are considered innovative.
	0 – 1 Marks	4 – 5 Marks	6 Marks	7-8 Marks
Research: Comparison [10 Marks]	- No comparison is included, nor is the comparison in a visual format. - Comparison in a visual format, but only includes either differences or similarities.	Comparison in a visual format with a good number of similarities and differences, but the presentation could be improved.	Comparison in a visual format with a good number of similarities and differences.	Comprehensive comparison that shows all the differences and similarities at a glance.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Research: List of features to include [6 Marks]	- No list of features included. - A very short list included with no motivation.	A good list of features is included, but with little or no motivation.	A good list of features is included with some motivation.	An excellent list of features with motivation included for why these features are desirable.
	0 – 2 Marks	3 – 4 Marks	5 Marks	6 Marks
Research: Conclusion [5 Marks]	- No conclusion included. - Conclusion does not link to the rest of the document.	Acceptable and links to the rest of the document.	Acceptable and links to the rest of the document. The link to the design is mentioned but not elaborated on.	An excellent conclusion that links to the document and the design.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks

PART 1 – Research				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard.
Planning and Design: Introduction and Conclusion [5 Marks]	- No introduction or conclusion included. - Either the introduction or conclusion is missing.	A good introduction and conclusion are included, as well as links to the rest of the document.	- A good introduction and conclusion are included, as well as links to the rest of the document - Mention is made of the research document.	An excellent introduction and conclusion that links to the research document as well as the content of the design document.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks

PART 1 – Planning and Design				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard.
Planning and Design: Overview of the app [10 Marks]	- No overview is included, and no innovative features are mentioned. - The overview needs more details.	A good overview is included with some innovative features.	A good overview is included with some innovative features and an attempt to link to the rest of the document.	An excellent overview that sets the stage for the rest of the document.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Planning and Design: Detailed list of requirements [10 Marks]	- No requirements are included, and the required features are missing. - Only required features are included with some details, but no additional features from the research are mentioned.	Required features and the student's requirements are included, but more details are needed in places.	Required features and the student's requirements are included, with more detail needed only in a few places.	An excellently detailed list of features that describes all the features, including the student's own research in detail.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Planning and Design: Creation of the REST API [10 Marks]	Little to no understanding of REST API	Some understanding of the implementation of making use of an API with the flow on the UML somewhat correct	Only UML created or detailed understanding of making use of a REST API to complete the application	- Good, detailed understanding of how to make use of a REST API to complete the application. - UML Diagram created
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Planning and Design: User interface design [10 Marks]	- No design is included, and only a very brief description with no mockups is provided. - Only mockups or reasonably detailed descriptions included, not both.	Mockups and descriptions included, but no diagrams showing navigation.	Mockups and descriptions included, and navigation diagrams included with some missing details.	Excellent mockups, together with descriptions and a diagram explaining the navigation.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks

PART 1 – Planning and Design				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard.
Planning and Design: Project Plan [5 Marks]	- No project plan included. - Project plan is included, but it is not logical.	Logical with some errors.	Logical with only minor errors.	Logical and fully detailed with no errors.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks

PART 2 – App Prototype Development				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard
The app runs on a mobile device. [5 Marks]	- App doesn't run at all. - The app runs, but lots of crashes happen.	- App runs with one or two errors or crashes. - Runs on the emulator	The app runs with very infrequent errors or crashes.	App runs without errors.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks
Feature: SSO sign-in [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: Settings menu that makes sense for the application [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Creation of the REST API [10 Marks]	- API not included. - API was created, but very buggy.	API working mostly with some bugs.	API working with only minor bugs.	API excellently created.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks

PART 2 – App Prototype Development				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard
Integration of your REST API in your app [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently integrated.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: User Defined 1 [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: User Defined 2 [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: User Defined 3 [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
User interface [10 Marks]	- Not user-friendly, confusing, and illogical. - Somewhat user-friendly with lots of mistakes in the design.	- Mostly user-friendly, with some mistakes in the design. - Somewhat consistent use of layout, fonts, and colours.	- Mostly user-friendly, with some attempt made at making the capture process easier for the user. - Mostly consistent use of layout, fonts, and colours.	- Excellent and user-friendly design. - Completely consistent use of layout, fonts, and colours.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks

PART 2 – App Prototype Development				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard
GitHub, Readme and automated testing [10 Marks]	Zip file added or file upload used for GitHub.	- No automated testing included. - GitHub repository submitted with limited commit history and limited evidence of collaboration. - No README.md file included	- Automated testing is included, but it is not working correctly. - GitHub repository submitted with good commit history (50 commits) - Evidence of collaboration (multiple contributors). - README.md file included but not formatted in Markdown.	- Automated testing is in place. - GitHub repository submitted with excellent commit history (50+ commits and well-structured commit messages). - Evidence of collaboration (multiple contributors). - README.md file included and perfectly formatted in Markdown.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Demonstration video [5 Marks]	- No demonstration video included. - Informal and unprepared, or not showing all the features or no voice-over included.	It is not completely professional, but all features are demonstrated.	Mostly professional with some issues, but all features are demonstrated.	Fully professional video showing all the required features in detail.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks

PORTFOLIO OF EVIDENCE (POE)				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard
The app runs on a mobile device. [5 Marks]	- App doesn't run at all. - The app runs, but lots of crashes happen.	- App runs with one or two errors or crashes. - Runs on the emulator	The app runs with very infrequent errors or crashes.	App runs without errors.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks
Feature: Biometric Authentication [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: Offline mode [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: Real-time Notification [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: Multi-language [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: User Defined 4 [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Feature: User Defined 5 [10 Marks]	- Feature not included. - Feature implemented but very buggy.	Feature working mostly with some bugs.	Feature working with only minor bugs.	Feature excellently implemented.

PORTFOLIO OF EVIDENCE (POE)				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
User interface [10 Marks]	- Not user-friendly, confusing, and illogical. - Somewhat user-friendly with lots of mistakes in the design.	- Mostly user-friendly, with some mistakes in the design. - Somewhat consistent use of layout, fonts, and colours.	- Mostly user-friendly, with some attempt made at making the capture process easier for the user. - Mostly consistent use of layout, fonts, and colours.	- Excellent and user-friendly design. - Completely consistent use of layout, fonts, and colours.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
The app has been prepared for publication in the Play Store. [5 Marks]	- No preparation has been done. - No evidence of preparation has been submitted. - The app is not ready for publication.	The minimum requirements for Play Store publication are met.	- The minimum requirements for Play Store publication are met. - Some additional material is included.	- The app is completely ready to be published to the Play Store. - All material, including screenshots for the Play Store, is included.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks
AI-use writeup [5 Marks]	Write-up not included.	Write-up included, but the details provided are vague.	Write-up does not include details of how AI was used and cited.	- A detailed write-up which includes information on how it was used and cited. - Mark may be awarded if AI is not used at all.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks

PORTFOLIO OF EVIDENCE (POE)				
Marking Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard
GitHub, Readme and automated testing [10 Marks]	Zip file added or file upload used for GitHub.	- No automated testing included. - GitHub repository submitted with limited commit history and limited evidence of collaboration. - No README.md file included	- Automated testing is included, but it is not working correctly. - GitHub repository submitted with good commit history (75 commits) - Evidence of collaboration (multiple contributors). - README.md file included but not formatted in Markdown.	- Automated testing is in place. - GitHub repository submitted with excellent commit history (75+ commits and well-structured commit messages). - Evidence of collaboration (multiple contributors). - README.md file included and perfectly formatted in Markdown.
	0 – 4 Marks	5 Marks	6 – 7 Marks	8 – 10 Marks
Demonstration video [5 Marks]	- No demonstration video included. - Informal and unprepared, or not showing all the features or no voice-over included.	It is not completely professional, but all features are demonstrated.	Mostly professional with some issues, but all features are demonstrated.	Fully professional video showing all the required features in detail.
	0 – 1 Marks	2 – 3 Marks	4 Marks	5 Marks
POE Total /100				

[TOTAL MARKS: 300]